

The Ultimate AWS Course Cheat Sheet (Lectures 1-13)

Here is a highly scannable cheat sheet organizing the core AWS services and concepts covered across your 13 lectures.

Core Infrastructure & Compute

Lecture	AWS Service / Concept	Primary Use Case & Key Features
L1	Cloud Concepts	Scalability, Elasticity, High Availability, Global Reach.

- | | L2 | IAM | Identity & Access Management. Securing AWS environment using Policies (JSON), Groups, and temporary Roles (Assume Role).
- | | L3 | Amazon EC2 | Virtual machines. Customized via AMIs and User Data (Bash scripts at boot).
- | | L3 | Amazon EBS | Network-attached block storage for EC2. Secured via Snapshots.
- | | L3 | Security Groups | Stateful virtual firewalls controlling inbound and outbound server traffic.
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Networking & Content Delivery

Lecture	AWS Service / Concept	Primary Use Case & Key Features
L4	Amazon VPC	Isolated virtual network (CIDR). Divided into Public (web) and Private (database) Subnets.

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- | | L5 | Amazon CloudFront | Global Content Delivery Network (CDN) using Edge Locations to cache data and reduce latency.
- | | L9 | Amazon Route 53 | Highly available DNS service. Routes traffic using policies like Simple, Alias, Failover, and Latency.
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Storage, Databases & Scaling

Lecture	AWS Service / Concept	Primary Use Case & Key Features
L5	Amazon S3	Infinite flat object storage. Uses Lifecycle Policies to move data between Standard, IA, and Glacier classes.

| | **L6 | Amazon RDS** | Managed relational databases. Uses Multi-AZ for Disaster Recovery and Read Replicas for performance scaling.

| | **L6 | Amazon DynamoDB** | Serverless, Key-Value NoSQL database delivering single-digit millisecond performance.

| | **L7 | Elastic Load Balancer** | Distributes traffic across EC2 instances. ALB for Layer 7 (HTTP) and NLB for Layer 4 (TCP/UDP).

| | **L7 | Auto Scaling Group (ASG)** | Automatically adjusts EC2 capacity using Target Tracking or Scheduled Scaling policies based on load.

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Monitoring, Decoupling & Serverless

Lecture	AWS Service / Concept	Primary Use Case & Key Features
L8	Amazon CloudWatch	Real-time performance monitoring. Collects metrics and triggers Alarms.

| | **L8 | AWS CloudTrail** | Auditing API calls (Who did what, when, and where).

| | **L8 | VPC Flow Logs** | Captures IP traffic metadata for network troubleshooting.

| | **L10 | AWS Lambda** | Serverless compute executing code in response to events (Event-Driven).

| | **L10 | Amazon SQS & SNS** | SQS for pull-based decoupling. SNS for push-based notifications.

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Architecture, Cost & Hybrid Migration

Lecture	AWS Service / Concept	Primary Use Case & Key Features
L11	Disaster Recovery	Strategies ranging from Backup & Restore, Pilot Light, Warm Standby, to Active/Active.

| | **L12 | Cost Tools & EC2 Pricing** | Cost Explorer and AWS Budgets. Utilizing On-Demand, Reserved, Savings Plans, and Spot Instances (up to 90% discount).

| | **L13 | Site-to-Site VPN** | Encrypted IPsec connection over public internet using VGW and CGW.

| | **L13 | AWS Direct Connect** | Dedicated physical fiber-optic connection bypassing public internet.

| | **L13 | Migration Strategies** | Rehost (Lift and Shift), Replatform (Lift, Tinker, Shift), and Refactor (Re-architect)